Ex:1
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Basic Networking Commands of Windows and Linux

AIM:

Networking Commands in Windows and Linux

Command 1:

window: ipconfig

Linux: ip addr

Purpose

Show your Ip address and networking details.

Example output

Wireless LAN adapter wifi:

IPv6 Address . . . . : 2406:7400:ce:4faf:1aec:2ffc
IPv4 Address . . . . : 192.168.88.9
Subnet Mask . . . . : 255.255.255.0

Command 2:

Window: ipconfig /all

Linux: ip address show

Purpose

Shows Complete network Configuration

Sample output

Windows IP configuration

Host Name . . . . . : DESKTOP-KNBPGCV
Primary Dns Suffix . . . . :
Node Type . . . . : Mixed
Ip Routing Enabled . No

Command 3:

window : Ping google.com

Linux: ping google.com

Purpose

checks if another device or website is reachable.

Sample output

Pinging google.com [142.250.77.110] with 32 bytes of data

Rely from 142.250.77.110 : bytes =32 time =5ms TTL = 119

Replay from 142.250.77.110 : bytes =32 time = 6ms TTL = 119

Ping statistics for 142.250.77.110 :

Packets : sent =4, Received = 4, Lost = 0 (0% Loss)

Approximate round trip times in milli - sec :

Minimum =5ms, max =8ms, Average = 6ms

Command 4:

window : tracert google.com

Linux: traceroute google.com

Purpose :

Shows the path packets take to a destination

Sample output

Tracing route to google.com [142.250.77.110]

over a maximum of 30 hops:

1	4ms	1ms	2ms	172.16.8.1
2	4ms	3ms	12ms	137.97.163.129
3	*	*	*	Request time out
4	17ms	5ms	5ms	172.16.12.122

Command 5:

Window : nslookup google.com

Linux : nslookup google.com

Purpose:

Finds the IP address of a domain

Sample output

Server : Unknown

Address : 172.16.8.1

Non-authoritative answer:

Name : google.com

Addresses : 2404:6800:4007:805::200e

142.250.67.46

Command 6:

window : netstat -an

Linux : netstat -tuln

Purpose

Display active network connections

Sample output

Active connections,

Proto	Local Address	Foreign Address	State
TCP	0.0.0.0.135	0.0.0.0.0	LISTENING
TCP	0.0.0.0.445	0.0.0.0.0	LISTENING
TCP	172.16.11.150.50756	4.213.25.240.443	ESTABLISHED
TCP	172.16.11.150.61909	184.28.109.178.443	TIME-WAIT
TCP	172.16.11.150.61918	74.226.60.120.443	CLOSE-WAIT

Command 7:

window : arp -a

Linux : arp -a

Purpose

Shows IP-to-MAC address mapping

Sample output

Interface: 172.16.11.150 --- 0x8

Internet Address	Physical Address	TYPE
172.16.8.1	7c-5a-1c-cf-be-45	dynamic
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Interface: 192.168.56.1 --- 0xb

Internet Address	Physical Address	TYPE
192.168.56.255	ff-ff-ff-ff-ff-ff	Static
224.0.0.22	01-00-5e-00-16	Static

Command 8:

window : hostname

Linux : hostname

purpose

Display the computer name

Sample output

window → DESKTOP-K[illegible]BRGICV

Linux → Proothop-pc